

Why Software Tests Fail and Hardware Attestation Wins

Are You Getting What You Pay For?

A technical, evidence-driven assessment for ML researchers, AI security practitioners, and cloud infrastructure engineers evaluating LLM API integrity.

black-box API

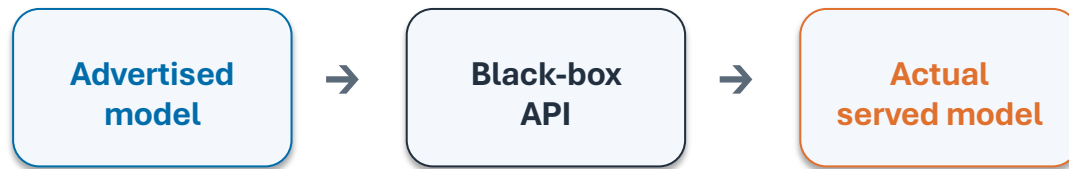
economic incentive

software audit gap

TEE attestation

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Commercial LLM APIs sell access to named models, but the black-box interface gives users no direct proof that the advertised weights are served.



Hypothesis test: distinguish $P_{\text{spec}}(y|x)$ from $P_{\text{actual}}(y|x)$ using only API samples.

Incentive: reduce GPU costs while maintaining advertised pricing.

Four covert substitution types

- 1 Smaller model**
e.g., 8B instead of 70B
- 2 Quantized variant**
FP8 / INT8 instead of FP16
- 3 Fine-tuned or updated**
same family, changed behavior
- 4 Different family**
entirely different architecture

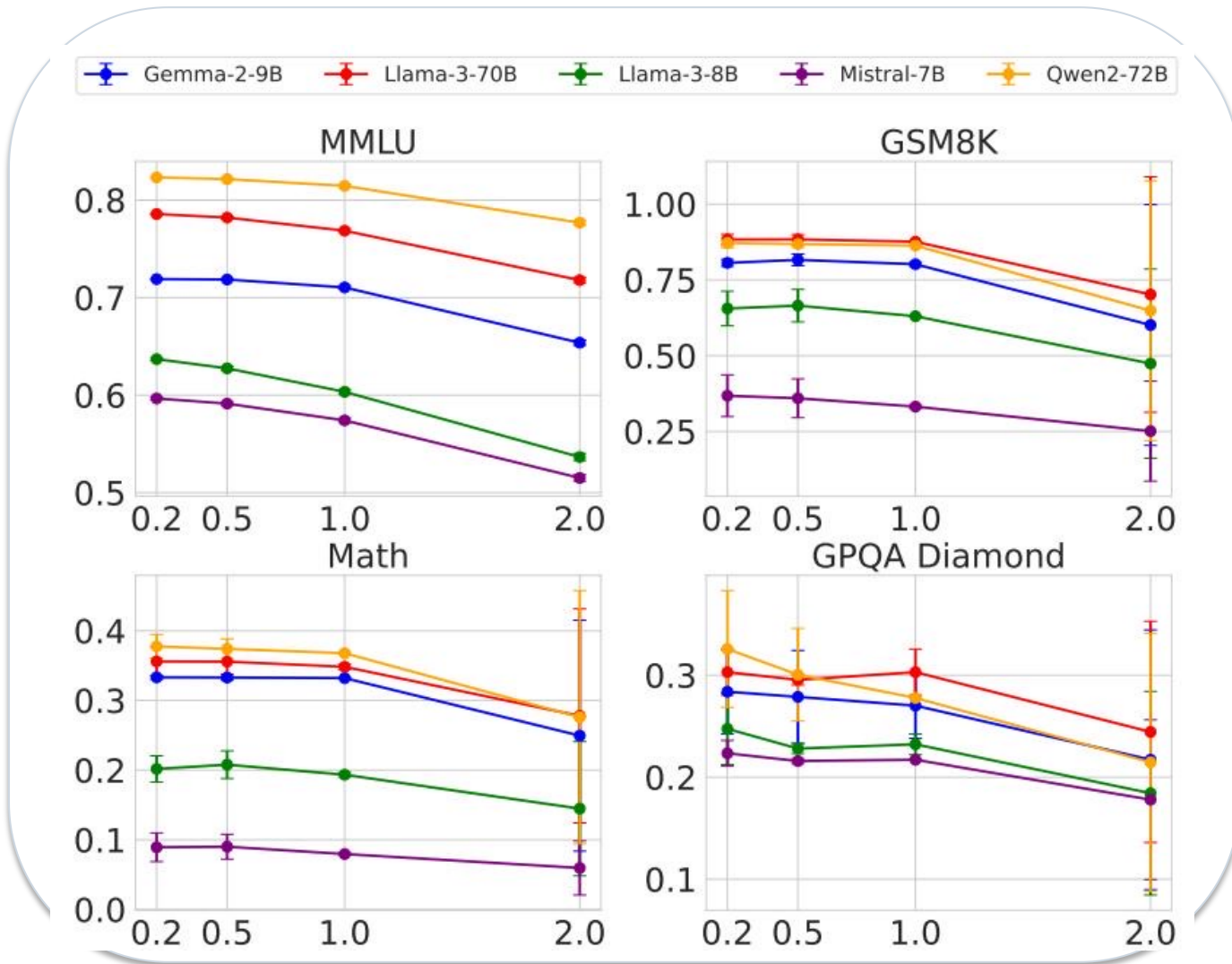
FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Five model families were evaluated in original precision vs. FP8 quantization across four benchmarks. Small deltas often sit inside reported error bars.

Model family	MMLU orig	MMLU FP8	GSM8K orig	GSM8K FP8	MATH orig	MATH FP8	GPQA orig	GPQA FP8
Llama-3-8B-Instruct	62.69 ±0.18	62.43 ±0.26	61.14 ±3.47	60.90 ±4.10	20.65 ±3.43	14.91 ±2.49	22.62 ±0.22	20.14 ±0.24
Llama-3-70B-Instruct	78.05 ±0.08	77.88 ±0.13	88.06 ±1.44	87.35 ±1.24	35.69 ±1.33	35.75 ±1.16	29.60 ±0.51	33.12 ±0.30
Gemma-2-9b-it	71.86 ±0.08	71.92 ±0.11	81.80 ±1.35	79.41 ±1.14	33.41 ±0.28	32.53 ±0.34	28.64 ±2.97	27.81 ±3.22
Qwen2-72B-Instruct	82.18 ±0.08	81.98 ±0.08	86.72 ±1.00	86.82 ±0.97	37.39 ±1.41	37.67 ±1.39	29.93 ±2.71	31.08 ±1.96
Mistral-7B-Instruct-v0.3	59.15 ±0.10	58.77 ±0.13	35.90 ±4.54	32.20 ±4.02	8.94 ±1.24	7.68 ±1.23	21.60 ±0.17	22.72 ±0.19

Interpretation: FP8 quantization can materially reduce serving costs while barely moving standard benchmark scores, making black-box detection statistically fragile.

Higher Temperature Amplifies Performance Gaps but Increases Noise



What the temperature sweep shows

- Benchmarks were run at temperatures 0.2, 0.5, 1.0, and 2.0.
- Higher temperatures degrade all models' performance across tasks.
- Noise increases and the original/substitute signal is further obscured.
- At temperature 2.0, gaps can narrow or become too noisy for robust statistical detection.

Figure reference: benchmark accuracy vs. temperature for five models across four tasks.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Token-level log probabilities look like a fingerprint, but production inference environments change the fingerprint even when the model is unchanged.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15

Software framework

transformers 4.40 vs 4.50; vLLM 0.6.2 vs 0.8.2

GPU hardware

A100 vs H100

Same model

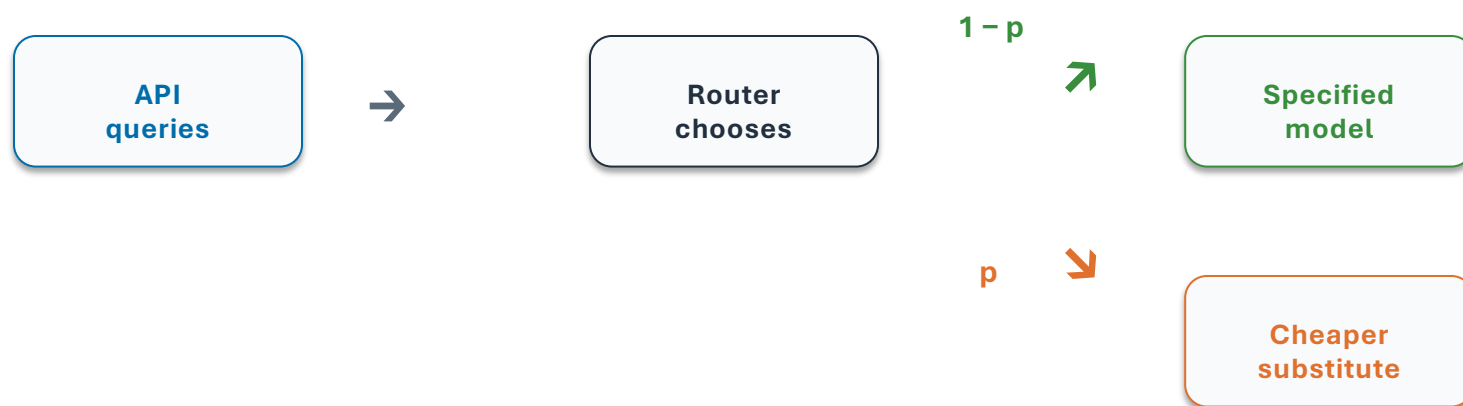
Llama-3-8B still produces varying log probabilities

Consequence: auditors cannot reliably distinguish legitimate infrastructure variation from actual model substitution.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

A provider can route only a fraction p of queries to a cheaper substitute. As p decreases, the observed API distribution converges to the specified model's distribution.

$$P_{\text{mix}}(y|x) = (1 - p) \cdot P_{\text{spec}}(y|x) + p \cdot P_{\text{sub}}(y|x)$$



Detection burden

$$p \rightarrow 0$$

P_{mix} approaches
 P_{spec}

Audits require exponentially more queries and become practically query-prohibitive.

Software-Only Auditing Methods Are Fundamentally Unreliable

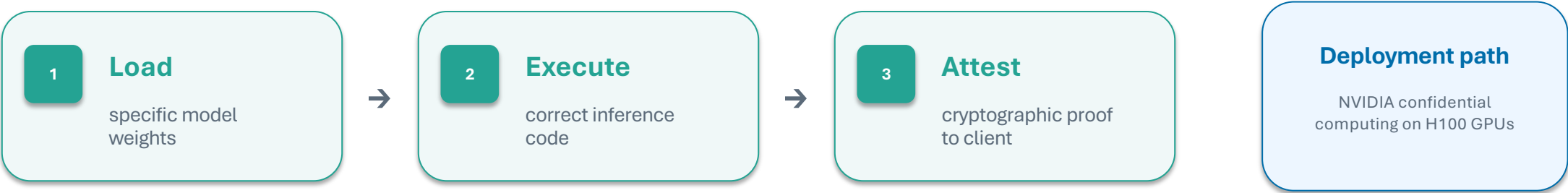
Each software-only path relies on observable outputs or metadata, but the paper’s evidence shows these signals are unstable or adversarially manipulable.

Method type	Weakness	Practical limitation
Text-based statistical tests (e.g., MMD)	Need too many queries and fail against subtle substitutions.	Impractically large query budgets; randomized substitution makes detection query-prohibitive.
Log-probability methods	Production inference nondeterminism changes token-level log probabilities.	Cannot separate a legitimate software/hardware deployment shift from true substitution.
Adversarial countermeasures	Benchmark evasion and randomized routing manipulate the audit surface.	Known benchmark prompts can be served by the genuine model while regular traffic receives substitutes.

Bottom line: software audits are observational; a motivated provider can exploit statistical ambiguity, infrastructure nondeterminism, and routing control.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

The proposed solution moves from output inference to hardware-backed proof: attest the code and model weights executing inside a secured enclave.



Security basis	statistical inference	cryptographic attestation
Robustness	evadable / noisy	hardware-rooted proof
Practicality	query-heavy	modest overhead; deployable today

Unlike ZKPs, which are computationally prohibitive for large models, TEEs are practical and deployable today.

Key Takeaways

Close the verification gap with hardware attestation.

1

Model substitution is economically motivated and practically undetectable via software-only audits.

2

FP8 quantization barely affects benchmarks, especially on MMLU and GSM8K.

3

Production nondeterminism defeats log-probability fingerprinting across software versions and GPU hardware.

4

Randomized routing and benchmark evasion undermine statistical detection and make audits query-prohibitive.

5

TEEs provide provable, efficient, robust model integrity guarantees and a practical deployment path.

Actionable standard: require verifiable inference attestations for commercial LLM APIs where model identity matters.